



ROOT CAUSE UNLOCKED · CONDITION KIT

The Mast Cell Root Cause *Kit*

Every test I order, why I order it, how to read it, and what actually changes when the number comes back

Dr. Adam Seay, DC

Root Cause Unlocked, Garner, NC · rootcauseunlocked.com · (919) 662-0520

The Mast Cell Root Cause Kit

Every test I order, why I order it, how to read it, and what actually changes when the number comes back

BEFORE YOU START

Contents

Read this part first

What you are holding, stated plainly

How I read a test, and the words I will and will not use

The thing no lab result outranks

Antihistamine response, which is not a laboratory test and belongs first anyway

Baseline serum tryptase, drawn on a day you feel well

Tryptase genotyping for hereditary alpha tryptasemia

The twenty-four hour urine mediator panel

Plasma histamine, and why I rarely order it

Serum diamine oxidase, read as a stratifier

AOC1 genotyping, and the sentence that keeps it honest

HNMT, the other exit, and why it matters for brain and airway symptoms

What I will not do with a methylation panel

Complete blood count with differential

Comprehensive metabolic panel

Ferritin, transferrin saturation, and CRP together

Vitamin B12 with methylmalonic acid

Iron and B12 in anyone on long-term acid suppression

25-OH vitamin D

hs-CRP

Zinc, copper, and ceruloplasmin

TSH, free T4, and free T3

Homocysteine, read correctly, including when it is low

Red blood cell folate

Celiac serology, and Helicobacter pylori, and autoimmune atrophic gastritis

Von Willebrand disease and a structural gynecological evaluation

A sleep apnea screen

The medication screen, which costs one question

Functional targets I publish

Where I publish no target, and say so

A request list you can bring

The question to ask before any panel is ordered

Working with Root Health

References

Read this part first

Where diet sits, and what it can actually do

The study you should read before you start

What the positive evidence actually shows

Lever one: preformed histamine in the food

Lever two: total amine load, not histamine alone

Lever three: your own gut bacteria, which is the lever nobody uses

Fermented foods: the conflict I am not going to smooth over

Storage and the cold chain, which is free and underrated

Alcohol, counselled for the right reason

The medication question, which is safety rather than comfort

The floor on breadth, and the end date on every restriction

Running a fair experiment

Realistic expectations

When to seek clinical review rather than more restriction

How you eat, which matters more than one more removed food

Working with Root Health

References

Read this part first

What you are buying, stated plainly

The gate: get the diet stable before you start

The cardinal rules that govern everything below

When you feel worse

Oral diamine oxidase enzyme, before amine-containing meals

Copper, at adequacy, and usually because zinc caused the problem

Vitamin C, at the dose that was actually studied

Gut lining repair, which is the actual lever

Probiotics, where the strain decides everything

Palmitoylethanolamide, which is the one with the best human data

Vitamin D, tested and to target

Quercetin, in the form that was actually studied

Luteolin, labeled for what it is

Butterbur, with a split verdict I will not smooth over

Boswellia, for the leukotriene arm, at modest confidence

Magnesium, kept for its real jobs

Omega-3, with the null result quoted

Riboflavin

B12, folate, B6 and betaine, as the four exits from homocysteine

The glutathione block

High-dose zinc as a stabilizer

Binders

Vitamin B6 for “DAO support”

Anything sold on the DAO-inhibiting drug list

Osmolality

Excipients

Speed

Where to find these, and what I learn

Working with Root Health

References

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient. Nothing in here diagnoses mast cell activation syndrome, hereditary alpha tryptasemia, mastocytosis, or histamine intolerance, and nothing in here is a cure for any of them.

Same-day medical attention, not a lab plan. Difficulty breathing, throat or tongue swelling, chest pain, fainting, confusion, or a reaction that keeps escalating after the trigger is gone. Those are emergencies. No panel result changes that, and none should delay that call.

One more thing before the tests. Symptoms attributed to mast cells and histamine also occur with true allergy, asthma, carcinoid and other neuroendocrine tumors, pheochromocytoma, thyroid disease, hereditary angioedema, autonomic disorders, anemia, medication reactions, and infection. Some of those are dangerous and treatable. A functional panel is not a substitute for that evaluation, and I would rather point you toward it than compete with it.

What you are holding, stated plainly

This is a guide to the tests I use, what each one can and cannot establish, and what I actually do when the result comes back. It is not a promise that testing will find an answer. In this particular field, several of the most heavily marketed tests are weak, one of the most useful ones is free, and the single most common testing error is reading a number against a population range when the criterion that matters is a change against your own baseline.

I would rather you understood that going in than found it out after four hundred dollars of panels.

Three things this guide will not do. It will not tell you that a number diagnoses you. It will not print a functional target where I do not have one I can defend. And it will not present a marketed test as validated when it is not.

How I read a test, and the words I will and will not use

I use these words deliberately, and it is worth knowing what each one means when you see it below.

Confirms. Reserved for tests genuinely validated for that exact use. Celiac serology followed by biopsy confirms celiac disease. A KIT D816V mutation on a marrow aspirate confirms clonality. Genotyping confirms hereditary alpha tryptasemia. Very little else in this guide earns that word.

Documents. A urinary mediator panel documents what was being excreted over the collection period. That is a real fact about a real window of time. It is not a diagnosis on its own.

Suggestive rather than diagnostic. Most functional markers live here. They shift my thinking and they do not settle a question.

Stratifies. This is the word I use most in this guide, and it is the honest one for genetics and for serum enzyme levels. It means the result tells me something about where you sit in a distribution, and therefore how to set expectations and what to try first. It does not tell me whether you are ill.

The thing no lab result outranks

Your pattern, meaning what happens, how long after what, in which organ systems, and whether an antihistamine has ever made any difference.

That last one is free and it outperforms several tests in this guide. It appears again in Part 1 for exactly that reason.

Part 1: The mediator tests

These are the tests that ask what is actually being released. They are the center of this workup.

Antihistamine response, which is not a laboratory test and belongs first anyway

If you have taken a standard second generation antihistamine, at an adequate dose, consistently, for long enough to judge it, and it did nothing at all, that result is diagnostic information.

Here is why. When a mast cell activates it releases histamine in the first seconds, then prostaglandin D2 and the cysteinyl leukotrienes over the following minutes, then cytokines over hours. An antihistamine blocks histamine receptors and nothing else. It touches neither prostaglandin D2 nor leukotrienes.

So a genuine non-response points away from histamine as your dominant mediator and toward the other two arms, which need a different test and a different class of intervention. That single observation redirects more of my plans than any panel below, and it costs nothing.

What changes when it comes back: if antihistamines never helped, the twenty-four hour urine panel moves up my list and the histamine-directed supplement stack moves down it.

Baseline serum tryptase, drawn on a day you feel well

Tryptase is stored in mast cell granules and a baseline value indexes roughly how many mast cells you have in total.

The reason it has to be drawn on a well day is that the value that matters clinically is the **change from your baseline during an event**, not the baseline itself. The criterion used in practice is a rise of 20 percent plus 2 ng/mL within about four hours of a reaction (PMID 34166845). Without a baseline, an acute value has nothing to be measured against and the criterion cannot be applied at all.

These are the two errors I see most.

The first is treating a single elevated tryptase as proof of a mast cell disorder. It is not, for the reason in the next section.

The second is treating a normal tryptase as proof that nothing is happening. Mast cells have two release modes. One is full degranulation, which dumps tryptase; the other is piecemeal or selective release, which is slow and partial and leaves no visible granule loss. Chronic daily mediator symptoms with a persistently normal tryptase is a recognized picture, not a contradiction.

Functional target: none published. Your laboratory's reference range governs the single value, and the change from your own baseline governs the interpretation.

What changes when it comes back: an elevated result sends me to tryptase genotyping before anything else. A very high result, or one with unexplained low blood counts, bone pain, or an enlarged liver or spleen, sends you to hematology rather than to me.

Tryptase genotyping for hereditary alpha tryptasemia

This is the test that almost nobody orders and that resolves most elevated tryptase results.

Roughly four to six percent of the general population carries extra copies of the alpha tryptase gene at TPSAB1. Those people run a persistently elevated baseline tryptase, commonly around 8 to 15 ng/mL, with no clonal disease and often no mast cell activation at all (PMID 27749843). Using genotype-defined individual reference ranges has been shown to prevent unnecessary procedures, including bone marrow evaluations that were never needed (PMID 36170795). And in people who do have systemic mastocytosis, the extra copies independently correlate with disease features, so the story is bigger than false positives (PMID 36309122).

This is a validated genetic test for a defined variant, so this is a place where the word **confirms** is correct.

What changes when it comes back: a positive result reframes your entire tryptase history and usually stops an escalating workup. A negative result with a persistently elevated tryptase raises the priority of specialist evaluation.

The twenty-four hour urine mediator panel

In my view this is the highest-yield test in this workup, because it is the one that separates the three arms from each other.

It measures three things at once (PMID 35346887):

- **11-beta-PGF2-alpha and 2,3-dinor-11-beta-PGF2-alpha**, which index prostaglandin D2.
- **LTE4**, the stable urinary form of the cysteinyl leukotrienes.
- **N-methylhistamine**, the validated urinary readout of histamine turnover (PMID 8859283, PMID 39745330).

Why this matters so much: a prostaglandin-dominant patient, a leukotriene-dominant patient and a histamine-dominant patient have overlapping symptoms and completely different answers. Treating all three identically with the same antihistamine-and-quercetin stack is the single most common structural error in this space, and this panel is what prevents it.

Practical notes. It is a timed collection, and the instructions matter more than usual. Some centers ask you to collect during or immediately after a reaction, which is where the information density is highest. Some of these analytes are not available at every reference laboratory and often require a specialty or academic center, so availability is a real constraint rather than a theoretical one.

Functional target: none published by me. The reporting laboratory's reference ranges govern.

What changes when it comes back: everything downstream. A leukotriene-dominant result makes the leukotriene conversation with your physician the priority. A prostaglandin-dominant result makes that a different conversation again, and one with a specific safety requirement attached, described in the supplement guide and the ebook. A histamine-dominant result means the diet and clearance work in Part 2 is aimed correctly.

Plasma histamine, and why I rarely order it

Plasma histamine did not move at all in healthy volunteers who took 75 mg of oral histamine and developed real symptoms from it (PMID 15603203). A blood level that does not respond when the molecule it measures is demonstrably active is not a useful readout of dietary or luminal histamine.

Whole blood histamine has a different problem: it largely tracks how many basophils you have, at roughly 1 to 2 picograms per basophil, and it moves when basophil counts move for reasons unrelated to your histamine handling (PMID 2477430). It is neither a methylation index nor a histamine burden index.

What changes when it comes back: very little, which is why I usually do not order it. It has a place drawn acutely during a documented reaction in a specialist setting. It has almost no place as a screening test.

Part 2: The clearance tests

These ask how well you get rid of histamine. Read this section knowing in advance that its tests are weaker than its marketing.

Serum diamine oxidase, read as a stratifier

Diamine oxidase is the enzyme that clears histamine outside your cells, concentrated in the villous tips of your small intestine.

The honest picture of the blood test: in patients with suspected histamine intolerance who underwent a placebo-controlled challenge, serum DAO tended lower in the confirmed cases but only at p equal to 0.08, and was highly variable among people who did not have the condition (PMID 37648152). It is not established that serum DAO reflects gut DAO (PMID 33921522). And it did not reliably move on a low histamine diet even in patients who carried the diagnosis (PMID 38769188).

Two confounders will invalidate the result outright if nobody catches them:

- **Recent heparin.** Most DAO is stored bound to the vessel wall and heparin releases it, so a post-heparin sample reads high for reasons that have nothing to do with your capacity (PMID 32985651, PMID 2110916).
- **Pregnancy.** The placenta expresses enormous quantities of DAO, so plasma DAO in pregnancy is uninterpretable for this purpose (PMID 18499706).

Functional target: none published by me. Your laboratory's range governs.

What changes when it comes back: a low value is the main reason I would run a trial of the oral enzyme, because benefit consistently concentrates in people selected that way. A normal value does not rule out a histamine problem and does not end the conversation.

AOC1 genotyping, and the sentence that keeps it honest

Variants in AOC1 that lower DAO activity are common. Over seventy percent of people of European ancestry carry at least one, and one specific allele lowers serum DAO with a clear gene-dose effect (PMID 17700358).

Then a study of 484 people found seven variants strongly associated with DAO activity, **and found that those variants were not associated with the histamine intolerance phenotype**, only with lower enzyme activity (PMID 21488903).

That sums it up in one line. Genotype identifies who has less enzyme. It does not identify who is ill.

What changes when it comes back: it stratifies. It helps me set expectations about how much headroom you have, and it makes the enzyme trial more attractive. It does not make a diagnosis, and I will not let it.

HNMT, the other exit, and why it matters for brain and airway symptoms

Histamine inside a cell is cleared by a different enzyme, histamine N-methyltransferase, which is cytosolic (PMID 7943261). Because the brain has essentially no diamine oxidase, the entire central nervous system depends on this route, and it depends on transporters to move histamine into the cell first (PMID 27679412, PMID 33641029). The airway depends on it too: blocking this enzyme potentiates histamine-induced contraction of isolated human bronchi, while blocking diamine oxidase does not (PMID 7943261).

Practically, that means if your symptoms are cognitive, neurological or airway-dominant, diet and a swallowed enzyme are acting on the wrong compartment, and that is useful to know before you spend a year on them.

A common HNMT variant lowers activity by raising the enzyme's requirement for its methyl donor (PMID 9547362, PMID 11566133), and complete loss of the gene causes a serious neurological condition, which is the cleanest proof the enzyme genuinely matters in humans (PMID 26206890). This is genuinely interesting biology. It is still not a diagnostic test, and I will not present it as one.

What I will not do with a methylation panel

The enzyme uses SAM as its methyl donor and leaves SAH behind, and SAH competitively inhibits the enzyme, so the variable that governs this route is the SAM to SAH ratio rather than homocysteine on its own (PMID 7188427).

The plasma ratio of SAM to SAH is the only validated methylation index, and a 2025 randomized trial lowering SAH by about 12 percent and raising the ratio by about 26 percent is the first demonstration that it can even be moved with nutrition (PMID 40883125). It is also not a test most people can obtain.

What I will not do is use whole blood histamine, or a low homocysteine, to label you an “overmethylator” or an “undermethylator.” Those constructs lack a published definition, a validated test, and a biomarker. Whole blood histamine cannot serve as one, per Part 1. And a low homocysteine has no validated reference interval that assigns it that meaning.

Homocysteine still earns its place in Part 3, read correctly, including when it is low.

Part 3: The baseline panel

None of these are mast cell tests. Every one of them changes what I do, and several of them are the reason someone feels better when the mediator work alone did not move them.

Complete blood count with differential

Basic, cheap, and load-bearing here. It screens anemia, and it also matters because unexplained low counts in any line is one of the findings that sends you to hematology rather than into another supplement round.

Functional target: none published. Your laboratory's range governs.

Comprehensive metabolic panel

Kidney and liver function, electrolytes, glucose. It exists partly to make sure nothing I suggest lands on an organ that cannot handle it, and partly because liver and kidney disease produce fatigue and symptom pictures that get attributed elsewhere.

Ferritin, transferrin saturation, and CRP together

This is the combination that matters, and ordering ferritin alone is the mistake.

Inflammation drives hepcidin, which blocks iron absorption and traps iron in storage, producing iron-restricted red cell production **with a normal or even high ferritin** (PMID 19786207). In a chronically activated patient, ferritin alone will mislead you. Transferrin saturation, soluble transferrin receptor where available, and a simultaneous CRP are what separate true depletion from sequestration.

And the finding that changes clinical practice most: fatigue responds to iron **without** anemia. A meta-analysis of six randomized trials in iron deficiency without anemia found a pooled effect size of 0.33 with zero heterogeneity (PMID 28625177), and the Cochrane review of 67 trials in 8,506 menstruating women found iron reduced anemia, raised hemoglobin, improved exercise performance and reduced symptomatic fatigue (PMID 27087396).

Which means **hemoglobin is the wrong screening test**. A normal hemoglobin does not clear you.

Functional target: ferritin 50 to 150 ng/mL. Transferrin saturation, serum iron and total iron binding capacity: none published by me, your laboratory governs.

What changes when it comes back: a low ferritin with a normal CRP is a repletion conversation, and a repletion conversation immediately raises the source question in Part 4. A normal or high ferritin with a high CRP and a low transferrin saturation is a different problem entirely and is not fixed by taking iron.

Vitamin B12 with methylmalonic acid

Order both. B12 alone is an unreliable marker and mean corpuscular volume is worse, because folate supplementation can normalize the red cell size while the B12 deficiency continues. Methylmalonic acid is the functional marker that catches what B12 alone misses.

There is a reason this sits in an MCAS guide specifically. Long-term H2 receptor antagonists carry an odds ratio of 1.25 for B12 deficiency and proton pump inhibitors 1.65, dose-dependently (PMID 24327038). H1 and H2 blockade is the backbone of conventional mast cell management, which means the standard treatment for the upstream problem is a documented mechanism for this deficiency. Nobody routinely monitors for it.

Functional target: B12 500 to 900 pg/mL. Methylmalonic acid: your laboratory's range governs.

Iron and B12 in anyone on long-term acid suppression

Worth its own line, because it follows directly from the paragraph above. The same drug classes carry odds ratios of 1.58 and 2.49 for iron deficiency, reaching 4.27 at high long-term proton pump inhibitor use (PMID 27890768).

This is not a reason to stop a medication that is working, and I will never tell you to stop one. It is a reason to check.

25-OH vitamin D

There is a real mechanistic reason this is in a mast cell guide rather than being generic advice. The vitamin D receptor physically interferes with a key mast cell activation signaling step, and mast cells spontaneously activate in a vitamin D deficient environment (PMID 27998003). In chronic urticaria, vitamin D as an add-on to antihistamine has randomized support (PMID 26097642).

Functional target: 40 to 80 ng/mL.

What changes when it comes back: a result below target is one of the few things in this guide where correcting the number has both a mechanism and a trial behind it at this node. I test rather than guess, and I retest rather than assume.

hs-CRP

A nonspecific screening marker. It tells me an inflammatory load exists somewhere. It never confirms anything, and it never tells me where the load is coming from.

It earns its place here for two specific jobs. One is interpreting ferritin, per above; the other is serving as the only practical proxy in a cash practice for the cytokine arm of mediator release, which otherwise lacks a marker, a baseline, and a retest.

Functional target: under 0.5 mg/L.

Zinc, copper, and ceruloplasmin

Copper is required by the diamine oxidase enzyme, twice over in the way the enzyme is built (PMID 20722416, PMID 27063138). But it behaves as a floor, not a dial: a frankly copper deficient diet lowers the enzyme, and supplementing copper well above adequacy changes nothing (PMID 10953681).

Which is why I test rather than supplement blind, and why I test all three together. The copper deficiency I actually find in practice is nearly always induced by a zinc supplement someone has been taking at a high dose for a long time.

Functional target: none published. Your laboratory's ranges govern.

TSH, free T4, and free T3

Thyroid disease produces fatigue, temperature intolerance, gut slowing, and heart rate changes that overlap this picture heavily. It is a cheap exclusion and a common finding.

Functional target: TSH 1.0 to 2.5 mIU/L. Free T4 and free T3: none published, your laboratory governs.

Homocysteine, read correctly, including when it is low

Homocysteine is a methylation cycle marker, not a cardiac number. When it is high, the useful move is to find the short input rather than to treat the number. The candidates are B12, folate, B6, or betaine, each of which serves a different exit from homocysteine (PMID 9398304, PMID 9681996).

And lower is not better. A homocysteine driven very low signals overmethylation and B vitamin intolerance, which is a real clinical picture and one worth catching rather than congratulating.

Functional target: 5 to 7 umol/L.

Red blood cell folate

Ordered alongside B12 when homocysteine is high, because it is one of the first inputs to check, not because it is a separate routine test.

Functional target: 600 to 1500 ng/mL.

Part 4: The tests that outrank the histamine story

If any of these apply to you, they take priority. They are more common, more treatable, and more consequential if missed. Skipping them to get to the interesting mediator tests is the error I most want you to avoid.

Celiac serology, and Helicobacter pylori, and autoimmune atrophic gastritis

Two separate reasons to run these.

First, iron. Celiac disease, autoimmune atrophic gastritis and H. pylori are the classic occult causes of iron deficiency that will not correct (PMID 15921373, PMID 24215034). Autoimmune atrophic gastritis deserves a second mention because it is the one diagnosis that explains both the iron arm and the B12 arm with a single lesion.

Second, and specific to this condition: diamine oxidase capacity is essentially a function of intact small intestinal villous surface. Post-heparin plasma DAO correlates with jejunal mucosal DAO at r equal to 0.84, drops sharply in active Crohn's disease in proportion to disease activity, and recovers in remission (PMID 2110916). In celiac disease, post-heparin DAO rose into the normal range within three months of a gluten free diet as the villi recovered (PMID 3121453), and reduced DAO shows up in both serum and biopsy tissue in childhood celiac disease (PMID 3088248). The Crohn's series did not reproduce a celiac deficit in its own subgroup, so this literature is not unanimous and I will say so.

But the direction is clear enough to act on: an undiagnosed mucosal disease is a ceiling on your histamine clearance that no supplement will lift.

Celiac serology followed by biopsy is a validated pathway, so **confirms** is the correct word here.

Von Willebrand disease and a structural gynecological evaluation

If heavy menstrual bleeding is part of your picture, this outranks mast cells by a wide margin.

Von Willebrand disease is a leading inherited cause of heavy menstrual bleeding, and women who have it carry high rates of heavy bleeding with iron deficiency and iron deficiency anemia (PMID 40697484, PMID 40955497). Fibroids and adenomyosis are the structural causes and they are found by imaging, not by a panel.

The reason this section sits in a mast cell guide needs stating plainly. Mast cells are present and degranulating in normal endometrium throughout the cycle, which is ordinary physiology rather than pathology (PMID 7492683), and histamine's predominant effect on human myometrium is contraction, which argues toward cramping rather than toward blood loss (PMID 1358393, PMID 15547545). Nobody has run a trial or a cohort, or even a case series, measuring menstrual blood loss in mast cell activation disease. The only quantitative human data linking endometrial mast cell counts to bleeding come from two small studies in intrauterine device users from 1970 and 1974, which is a foreign body model (PMID 4193675).

So it is plausible and unproven, and not where I would look first. The bleeding itself is the treatable target, and stopping the loss at the source is what makes iron repletion hold.

A sleep apnea screen

Obstructive sleep apnea causes intermittent low oxygen. It is not caused by anemia, and on most versions of the popular histamine map the arrow points the wrong way.

In 527 patients with sleep apnea, over 98 percent had **no** elevated hemoglobin, because inflammation-driven hepcidin limits iron availability for red cell production and newly formed cells are destroyed (PMID 37350302). Untreated sleep apnea therefore contributes to iron-restricted erythropoiesis and produces exactly the fatigue, cognitive and headache picture being attributed to mediator burden.

If you snore, wake unrefreshed, or wake with a headache, get screened. This is one of the highest-yield items in this entire guide and it is not a laboratory test.

The medication screen, which costs one question

Two things I ask about specifically.

Monoamine oxidase inhibitors. Phenelzine, tranylcypromine, isocarboxazid, high-dose selegiline, and linezolid. Combined with a tyramine-rich meal, these can produce a hypertensive crisis (PMID 3525654, PMID 17192761). Tyramine shares a food list with histamine and shares nothing else mechanistically, which is exactly why this gets missed in a histamine conversation. This is safety, not comfort.

MRGPRX2 agonists. Opioids, fluoroquinolone antibiotics, vancomycin, and neuromuscular blockers used in anesthesia activate mast cells directly through a receptor with no IgE involvement at all (PMID 25517090). If you have reacted to one of these and been told it was not a true allergy, both statements can be correct: it was not IgE-mediated, and it was completely real.

And one thing I will tell you **not** to worry about. The circulating internet lists of “drugs that block DAO” are largely unsupported. Tested directly, ibuprofen, aspirin, paracetamol, sumatriptan and zolmitriptan did not inhibit the enzyme. Only naproxen lowered DAO protein, and without affecting activity (PMID 38068554). Ethanol, acetaldehyde and disulfiram do not inhibit human DAO either, and the apparent disulfiram effect turned out to be an assay artifact (PMID 27401969).

Part 5: The numbers

Functional targets I publish

These are my clinical targets, used at Root Health. They are narrower than the standard laboratory range on purpose, and they are not validated diagnostic thresholds. Where the standard column says your laboratory governs, it means exactly that.

Marker	Units	Functional target	Approximate standard range
Vitamin D (25-OH)	ng/mL	40 to 80	20 to 100
Ferritin	ng/mL	50 to 150	25 to 300
Homocysteine	umol/L	5 to 7	under 10 to 15, varies by lab
hs-CRP	mg/L	under 0.5	under 1.0
TSH	mIU/L	1.0 to 2.5	your laboratory's range governs
Vitamin B12	pg/mL	500 to 900	200 to 1100
Folate (RBC)	ng/mL	600 to 1500	400 to 2000

Where I publish no target, and say so

For the following I have no functional target I can defend, so I print none rather than invent one. Your laboratory's reference range and your clinician's interpretation govern each of them.

Serum tryptase. Twenty-four hour urine N-methylhistamine, 11-beta-PGF2-alpha and its dinor metabolite, and LTE4. Plasma and whole blood histamine. Serum diamine oxidase. AOC1 and HNMT genotype. Serum zinc, serum copper, and ceruloplasmin. Serum iron, total iron binding capacity, and transferrin saturation. Methylmalonic acid. Free T4 and free T3.

That list is longer than the one above it, and that is the honest shape of this field. A guide that printed a confident functional target for a urinary mediator would be telling you something nobody knows.

Part 6: Reading your results as a pattern

Individual numbers rarely settle anything here. Patterns do. These are the four I look for most.

The clearance pattern. Symptoms clustering three to six hours after amine-heavy meals, worse with leftovers and aged food, better on a structured elimination, with a low or low-normal serum DAO and possibly an AOC1 variant. Antihistamines may help partially. This picture responds to diet, mucosal repair, and sometimes the oral enzyme, and it does not need a mast cell stabilizer.

The activation pattern. Multi-system symptoms, physical and non-food triggers, reactions to medications and excipients, flushing, and a tryptase that rises during events even if the baseline is normal. This is where the mediator panel earns its money.

The wrong-mediator pattern. Multi-system symptoms that read exactly like mast cell activation, and antihistamines that did nothing at any dose. This is the picture that needs a urine panel before anything else, because it is probably being driven by prostaglandin D2 or the leukotrienes, and every histamine-directed thing tried so far has been aimed at the wrong target.

The depletion pattern. Exhaustion, fog, dizziness on standing, hair loss, and exercise intolerance, with a low ferritin, sometimes a normal hemoglobin, and often heavy periods or long-term acid suppression in the history. This is the one that is genuinely fixable, that gets attributed to mediator burden, and that I would want treated first, because a person who is iron depleted cannot tell you whether anything else is working.

Part 7: The conversation with your doctor

A request list you can bring

This is written to be handed over. Not every item will be appropriate for you, and your physician may have good reasons to decline some of them. That is a conversation, not a defeat.

Mediator tests

- Serum tryptase, drawn on a symptom-free day, as a baseline
- Serum tryptase drawn within about four hours of a reaction, if one occurs
- Tryptase genotyping for hereditary alpha tryptasemia, if baseline tryptase is elevated
- Twenty-four hour urine mediator panel: N-methylhistamine, 11-beta-PGF2-alpha and 2,3-dinor-11-beta-PGF2-alpha, and LTE4

Baseline

- Complete blood count with differential
- Comprehensive metabolic panel
- Ferritin, transferrin saturation, and hs-CRP together
- Vitamin B12 with methylmalonic acid
- 25-OH vitamin D
- TSH, free T4, free T3
- Homocysteine, and red cell folate if homocysteine is elevated
- Serum zinc, serum copper, and ceruloplasmin

Differential, if the relevant history applies

- Celiac serology
- Helicobacter pylori testing, and evaluation for autoimmune atrophic gastritis
- Von Willebrand testing and a structural evaluation, if heavy menstrual bleeding
- A sleep study, if snoring, unrefreshing sleep, or morning headache

Questions worth asking out loud

- If my tryptase is elevated, has hereditary alpha tryptasemia been excluded
- Am I a candidate for an emergency action plan and an epinephrine prescription
- Should I be seen by allergy and immunology, or by hematology
- I take acid suppression long-term: should my iron and B12 be monitored

The question to ask before any panel is ordered

What will change depending on the result.

If neither answer changes anything you would do, the test is expensive curiosity. That question has saved my patients more money than any single recommendation in this guide, and I would rather you asked it of me than of anyone else.

Working with Root Health

If you want help reading a panel you already have, or deciding which of these is worth running, the next step is a conversation. The Root Discovery Session is free, and it exists exactly for that. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere. In this condition it does real work, because the timed pattern it captures outperforms several of the tests in this guide.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

This guide is educational. It is not medical advice, diagnosis, or treatment, and reading it does not create a doctor and patient relationship.

References

1. Valent P, et al. Selecting the right criteria and proper classification to diagnose mast cell activation syndromes: a critical review. *J Allergy Clin Immunol Pract.* 2021. PMID 34166845.
2. Lyons JJ, et al. Elevated basal serum tryptase identifies a multisystem disorder associated with increased TPSAB1 copy number. *Nat Genet.* 2016. PMID 27749843.
3. Lyons JJ, et al. Genetically defined individual reference ranges for tryptase limit unnecessary procedures and unmask myeloid neoplasms. *Blood Adv.* 2023. PMID 36170795.
4. Greiner G, et al. Disease correlates and clinical relevance of hereditary alpha-tryptasemia in patients with systemic mastocytosis. *J Allergy Clin Immunol.* 2023. PMID 36309122.
5. Ravi A, et al. Nontryptase urinary and hematologic biomarkers of mast cell expansion and mast cell activation: status 2022. *J Allergy Clin Immunol Pract.* 2022. PMID 35346887.
6. Value of urinary N-methylhistamine measurements in childhood mastocytosis. *J Am Acad Dermatol.* 1996. PMID 8859283.
7. Biomarkers in the diagnosis of mast cell activation. *Curr Opin Allergy Clin Immunol.* 2025. PMID 39745330.
8. Wohrl S, et al. Histamine intolerance-like symptoms in healthy volunteers after oral provocation with liquid histamine. *Allergy Asthma Proc.* 2004. PMID 15603203.
9. Comparison between number of basophils, blood histamine, and histamine release. *J Allergy Clin Immunol.* 1989. PMID 2477430.
10. Placebo-controlled histamine challenge disproves suspicion of histamine intolerance. *J Allergy Clin Immunol Pract.* 2023. PMID 37648152.
11. Schnedl WJ, Enko D. Histamine intolerance originates in the gut. *Nutrients.* 2021. PMID 33921522.
12. Measurement of diamine oxidase during low-histamine or ordinary diet in patients with histamine intolerance. *Eur J Clin Nutr.* 2024. PMID 38769188.
13. Human diamine oxidase cellular binding and internalization in vitro and rapid clearance in vivo. *Glycobiology.* 2021. PMID 32985651.
14. Wollin A, et al. Postheparin plasma diamine oxidase in health and intestinal disease. *Gastroenterology.* 1990. PMID 2110916.
15. Maintz L, et al. Effects of histamine and diamine oxidase activities on pregnancy: a critical review. *Hum Reprod Update.* 2008. PMID 18499706.
16. Maintz L, et al. Genetic variability of human diamine oxidase: occurrence of three nonsynonymous polymorphisms. *Pharmacogenet Genomics.* 2007. PMID 17700358.
17. Maintz L, et al. Association of single nucleotide polymorphisms in the diamine oxidase gene with diamine oxidase serum activities. *Allergy.* 2011. PMID 21488903.
18. Yamauchi K, et al. Structure and function of human histamine N-methyltransferase: critical enzyme in histamine metabolism in airway. *Am J Physiol.* 1994. PMID 7943261.
19. Histamine clearance through polyspecific transporters in the brain. *Handb Exp Pharmacol.* 2017. PMID 27679412.
20. Organic cation transporters in brain histamine clearance. *Handb Exp Pharmacol.* 2021. PMID 33641029.

21. Preuss CV, et al. Human histamine N-methyltransferase pharmacogenetics: common genetic polymorphisms that alter activity. *Mol Pharmacol*. 1998. PMID 9547362.
22. Horton JR, et al. Two polymorphic forms of human histamine methyltransferase: structural, thermal, and kinetic comparisons. *Structure*. 2001. PMID 11566133.
23. Heidari A, et al. Mutations in the histamine N-methyltransferase gene, HNMT, are associated with nonsyndromic autosomal recessive intellectual disability. *Hum Mol Genet*. 2015. PMID 26206890.
24. Francis DM, et al. The kinetic properties and reaction mechanism of histamine methyltransferase from human skin. *Biochem J*. 1980. PMID 7188427.
25. Lowering plasma S-adenosylhomocysteine in healthy adults with elevated SAH and normal homocysteine using nutritional supplementation. *Nutr Metab Cardiovasc Dis*. 2025. PMID 40883125.
26. Ganz T, Nemeth E. Iron sequestration and anemia of inflammation. *Semin Hematol*. 2009. PMID 19786207.
27. Houston BL, et al. Iron deficiency without anaemia is a potential cause of fatigue: meta-analyses of randomised controlled trials. *Br J Nutr*. 2017. PMID 28625177.
28. Low MSY, et al. Daily iron supplementation for improving anaemia, iron status and health in menstruating women. *Cochrane Database Syst Rev*. 2016. PMID 27087396.
29. Lam JR, et al. Proton pump inhibitor and histamine 2 receptor antagonist use and vitamin B12 deficiency. *JAMA*. 2013. PMID 24327038.
30. Tran-Duy A, et al. Proton pump inhibitor and histamine-2 receptor antagonist use and iron deficiency. *Gastroenterology*. 2017. PMID 27890768.
31. Liu ZQ, et al. Vitamin D contributes to mast cell stabilization. *Allergy*. 2017. PMID 27998003.
32. Rasool R, et al. Chronic urticaria merits serum vitamin D evaluation and supplementation: a randomized case control study. *World Allergy Organ J*. 2015. PMID 26097642.
33. Correlation of active site metal content in human diamine oxidase with trihydroxyphenylalanine quinone cofactor biogenesis. *Biochemistry*. 2010. PMID 20722416.
34. Characterization of recombinant human diamine oxidase produced in CHO cells. *J Biotechnol*. 2016. PMID 27063138.
35. Response of diamine oxidase and other plasma copper biomarkers to various dietary copper intakes in the rat. *Br J Nutr*. 2000. PMID 10953681.
36. Goulding CW, Matthews RG. Cobalamin-dependent methionine synthase: involvement of zinc in homocysteine activation. *Biochemistry*. 1997. PMID 9398304.
37. Millian NS, Garrow TA. Human betaine-homocysteine methyltransferase is a zinc metalloenzyme. *Arch Biochem Biophys*. 1998. PMID 9681996.
38. Hershko C, et al. Role of autoimmune gastritis, *Helicobacter pylori* and celiac disease in refractory or unexplained iron deficiency anemia. *Haematologica*. 2005. PMID 15921373.
39. Camaschella C. How I treat unexplained refractory iron deficiency anemia. *Blood*. 2014. PMID 24215034.
40. Postheparin plasma diamine oxidase increases in patients with coeliac disease during gluten free diet. *Gut*. 1987. PMID 3121453.
41. Diamine oxidase in serum and small intestinal biopsy tissue in childhood celiac disease. *J Pediatr Gastroenterol Nutr*. 1986. PMID 3088248.

42. Prevalence of heavy menstrual bleeding, iron deficiency, iron deficiency anemia, and treatment in women with von Willebrand disease. *Res Pract Thromb Haemost.* 2025. PMID 40697484.
43. Prevalence and impact of heavy menstrual bleeding in women with von Willebrand disease across age groups. *Haemophilia.* 2025. PMID 40955497.
44. Jeziorska M, et al. Mast cell and eosinophil distribution and activation in human endometrium throughout the menstrual cycle. *Biol Reprod.* 1995. PMID 7492683.
45. Effect of histamine and histamine analogues on human isolated myometrial strips. *Br J Pharmacol.* 1992. PMID 1358393.
46. Degranulation of uterine mast cell modifies contractility of isolated myometrium from pregnant women. *Am J Obstet Gynecol.* 2004. PMID 15547545.
47. Sheppard BL, Bonnar J. The relationship between endometrial mast cell count and bleeding in women following insertion of an intrauterine device. *Am J Obstet Gynecol.* 1970. PMID 4193675.
48. Increased blood reactive oxygen species and hepcidin in obstructive sleep apnea precludes expected erythrocytosis. *Am J Hematol.* 2023. PMID 37350302.
49. McCabe BJ. Dietary tyramine and other pressor amines in MAOI regimens: a review. *J Am Diet Assoc.* 1986. PMID 3525654.
50. Azzaro AJ, et al. Effects of a tyramine-enriched meal on blood pressure response in healthy male volunteers treated with selegiline transdermal system. *CNS Spectr.* 2007. PMID 17192761.
51. McNeil BD, et al. Identification of a mast-cell-specific receptor crucial for pseudo-allergic drug reactions. *Nature.* 2015. PMID 25517090.
52. Diamine oxidase interactions with anti-inflammatory and anti-migraine medicines in the treatment of migraine. *J Clin Med.* 2023. PMID 38068554.
53. Recombinant human diamine oxidase activity is not inhibited by ethanol, acetaldehyde, disulfiram, diethyldithiocarbamate. *Alcohol.* 2016. PMID 27401969.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose anything, and reading it does not make you my patient.

Anaphylaxis is a medical emergency. Difficulty breathing, throat or tongue swelling, a drop in blood pressure, or fainting means calling 911. If you have had a reaction like that, an emergency action plan and an epinephrine prescription from a physician come before any dietary plan.

If you have a diagnosed IgE food allergy, this guide does not change one thing about it. Avoidance of a true allergen is absolute and is not a matter of load or timing.

Do not run an elimination diet in pregnancy, in a child, in anyone with an eating disorder history, or in anyone already underweight without clinical supervision. Narrowing an already narrow diet is a real risk, and in this condition it is a risk with a specific mechanism attached, covered below.

Where diet sits, and what it can actually do

Diet is the highest-yield thing you can do in a histamine-dominant picture, and it is close to useless in a picture driven by other mediators.

That distinction is worth knowing before you start. When a mast cell activates it releases histamine in the first seconds, then prostaglandin D2 and the cysteinyl leukotrienes over the following minutes. Diet acts on the histamine arriving from outside plus the histamine your gut bacteria make. It does not act on the other two arms at all.

So a person whose urine mediator panel is leukotriene-dominant can run a perfect low-histamine diet for six months and change nothing. And a person whose problem is genuinely load-versus-clearance can get most of the way home in three weeks.

I would rather you knew which one you are before you reorganize your kitchen.

The study you should read before you start

This is the most important thing in this guide, and it is uncomfortable.

In fifty-nine patients with suspected histamine intolerance, a single-blind placebo-controlled oral histamine challenge **excluded** the diagnosis in fifty of them, roughly 85 percent. Thirty-seven of the fifty-nine, about 63 percent, reacted to the **placebo**. Only about 7 percent had objective symptoms to histamine and not to placebo (PMID 37648152).

And three quarters of those patients had already reported improving on a low histamine diet.

Two things follow, and I want both of them held at once.

First: run the diet anyway. It is low risk, it is time-limited, and it is the fastest way to find out whether load is your problem. There is real positive human evidence for it, covered below.

Second: do not treat feeling better as proof of the mechanism. Expectation is powerful in this condition, and the trial design that separates the two is not one you can run at home. Hold the conclusion loosely, keep the end date, and let the reintroduction phase test what the elimination phase suggested.

Calling it hedging misses the point. It is the difference between a three-week experiment and a three-year restriction.

What the positive evidence actually shows

Three studies carry the weight here.

Three weeks of a low-histamine diet in fifty-six patients with chronic spontaneous urticaria and gastrointestinal symptoms, with 61 percent reaching the predefined endpoint (PMID 27624921). Four weeks in twenty-two adults with urticaria, with symptom scores and plasma histamine both falling (PMID 29606813). And a randomized crossover in eighteen diagnosed patients showing significant gastrointestinal and skin reductions (PMID 38769188).

The limits, stated plainly. Two of the three are open-label with no placebo arm, which matters a great deal given the placebo rate above. And **diamine oxidase activity did not move in any of them**. That last point is the one people miss: this is a symptom test, not an enzyme test. Improving on the diet does not mean your enzyme changed, or that it was ever the problem.

Which is why I prescribe this as a time-limited diagnostic and therapeutic trial and never as a permanent way of eating.

Lever one: preformed histamine in the food

Food is a bioreactor. Fresh protein carries histidine mostly bound up inside protein and is a poor histamine source. Ripening, aging, fermentation, and bacterial breakdown liberate free histidine, and then bacterial enzymes turn it into histamine.

Two properties make this practical rather than theoretical.

Histamine is heat stable. Cooking kills the organism and leaves the histamine sitting there. Scombroid poisoning, where a badly handled fish produces the full syndrome in an otherwise healthy person, is the proof that dietary histamine alone is pharmacologically sufficient (PMID 25876709).

Refrigeration does not stop it. Cold-tolerant bacteria keep producing histamine at 1 to 3 degrees C. Vacuum-packed tuna held near 2 degrees C reached over 5,000 mg/kg, and modified-atmosphere packaging did not prevent it (PMID 15925710, PMID 18845350).

The practical categories, in the order they usually matter:

- **Aged, cured and smoked meats and fish.** The highest-yield category and the one to remove first.
- **Aged cheeses**, with fresher soft cheeses lower.
- **Fermented foods**, which get their own section below because the picture is genuinely mixed.
- **Alcohol**, especially wine and beer.
- **Leftovers**, which is the category people underestimate most.
- **Canned and jarred fish**, and anything that has spent time in a supply chain.

Lever two: total amine load, not histamine alone

Diamine oxidase is saturable and it is not selective. Putrescine, cadaverine and tyramine present in the same meal compete with histamine for the same active site and slow its degradation (PMID 35694170, PMID 40865824).

That single fact resolves the thing that frustrates people most about low-histamine food lists: the lists do not agree with each other, and many of the foods on them contain almost no histamine.

Across ten published low-histamine diets, only about **32 percent** of the excluded foods were excluded on histamine content (PMID 33919293). Bananas and citrus are the classic examples. They are not on those lists for their histamine.

The useful shift is from obeying a list to reasoning about a meal. Ask what the total amine load of this plate is, not whether one item appears on someone's chart. A meal of fresh food with one moderate-amine component is a very different proposition from a meal where four moderate items stack.

Lever three: your own gut bacteria, which is the lever nobody uses

Some of your histamine is being made after the food is eaten, by your own microbes.

Fecal bacteria from irritable bowel syndrome patients with high urinary histamine produce large amounts of histamine in the laboratory, and a specific organism carrying a histamine-producing gene variant was identified as a major producer and was over-represented across three independent human cohorts (PMID 35895832). Case-control work in histamine intolerance finds a consistent pattern: more of certain histamine-producing organisms, fewer of the protective ones (PMID 35565742, PMID 30552302).

Here is the part that makes this actionable. In a randomized trial, three weeks of a low fermentable carbohydrate diet versus a high one reduced urinary histamine roughly **eightfold** (PMID 26976734).

That is the only randomized trial in this entire literature that used a histamine biomarker as an outcome. And it works by a completely different mechanism from a low-histamine diet: it starves the bacterial enzyme of the substrate it needs, rather than removing preformed histamine from your plate.

Who I use it in. Anyone whose picture is gut-dominant, with bloating, cramping, or altered bowel habit alongside the histamine symptoms. It is also worth trying in someone who ran a careful low-histamine diet and got nowhere, because it is a different lever entirely and a null result on one says nothing about the other.

The same rule applies. Time-limited, structured, with reintroduction. A low fermentable carbohydrate diet run indefinitely has its own costs to the microbiome.

Fermented foods: the conflict I am not going to smooth over

All ten published low-histamine diets exclude fermented foods, on measured amine content (PMID 33919293).

Against that, a seventeen week randomized trial of six servings of fermented foods per day raised microbiota diversity and lowered nineteen inflammatory proteins in healthy adults. It did not measure histamine at all (PMID 34256014).

Those two findings point in opposite directions and both are real. My reading: fermented food is probably good for a general population and probably a poor idea during an active elimination phase in someone whose amine load is the problem.

This matters practically for a reason that catches people out. Gut repair programs quietly push fermented food, and a person can end up running a low-histamine diet and a fermented food protocol at the same time without noticing the contradiction. Decide deliberately. Reintroduction, later and on purpose, is where I put it.

Storage and the cold chain, which is free and underrated

For some people this section does more than any supplement they will ever buy.

- **Buy fish fresh and eat it the same day.** If you cannot, buy it frozen at sea rather than “fresh” that has been on a counter.
- **Freeze leftovers immediately**, do not refrigerate them. Cold-tolerant bacteria keep working in a domestic refrigerator (PMID 18845350).
- **Cook and eat, or cook and freeze.** The forty-eight hours in a container in the fridge is where the amine load builds.
- **Reheating does nothing.** Histamine is heat stable (PMID 25876709). You cannot cook it out.
- **Ask how long it has been at the counter.** For fish specifically, that question is more informative than the species.
- **Shorter supply chains beat longer ones**, which is most of what people are actually noticing when they say fresh food agrees with them.

Nothing in this section costs money and nothing in it removes a food from your diet. It is the first thing I would change.

Alcohol, counselled for the right reason

Alcohol belongs in this conversation and the usual reason given for it is wrong.

The claim that ethanol and acetaldehyde inhibit diamine oxidase does not survive testing. Neither compound, nor disulfiram, inhibited recombinant human DAO at physiologically relevant concentrations, and the apparent disulfiram effect turned out to be an assay artifact (PMID 27401969).

Alcohol earns its place on two other grounds: the amine content of wine and beer, and direct mast cell provocation. That is enough. It does not need a mechanism that is not there.

The medication question, which is safety rather than comfort

One screen matters here and it is not optional.

If you take a monoamine oxidase inhibitor, phenelzine, tranylcypromine, isocarboxazid, high-dose selegiline, or linezolid, a tyramine-rich meal can produce a hypertensive crisis (PMID 3525654, PMID 17192761).

Tyramine shares a food list with histamine and shares nothing else mechanistically, which is exactly why this gets missed inside a histamine conversation. If any of those apply to you, talk to your prescriber before you touch aged or fermented foods.

And one thing not to worry about: the circulating lists of “drugs that block DAO” are largely folklore. Tested directly, ibuprofen, aspirin, paracetamol, sumatriptan and zolmitriptan did not inhibit the enzyme. Only naproxen lowered DAO protein, and without affecting activity (PMID 38068554).

The floor on breadth, and the end date on every restriction

Every food list on the internet leaves this out, and I care about it more than anything else here.

Diamine oxidase capacity is essentially a function of healthy small intestinal lining. The enzyme lives in the villous tips, so how much enzyme you have tracks how much intact absorptive surface you have. Post-heparin plasma diamine oxidase correlates with jejunal mucosal diamine oxidase at r equal to 0.84 (PMID 2110916).

Now the finding that should shape how you run this. In that same work, post-heparin DAO was 4.5 during total parenteral nutrition against 35.9 in controls, and it **reversed on resumption of oral feeding** (PMID 2110916). Food in the lumen maintains the cell mass that is the enzyme.

Intravenous feeding is a far more extreme insult than an elimination diet, and I am not going to claim this proves elimination diets damage your enzyme. It does not. But the direction of the finding should concern anyone who has been eating the same fifteen foods for a year, because the plausible reading is that a prolonged, very narrow diet may be eroding the very clearance capacity being blamed for the problem.

So two rules, and they are not negotiable in my plans:

Every restriction has an end date, set on the day it starts.

Every plan has a floor on dietary breadth. If the number of foods you can eat is going down month after month, the plan is failing regardless of how you feel this week.

Running a fair experiment

Weeks 1 to 3, the elimination. Pick one lever, not three. Low histamine if your pattern is aged-food and leftover timed; low fermentable carbohydrate if it is gut-dominant. Running both at once tells you nothing about either and narrows your diet twice as fast.

Keep the timed diary. Use four columns, for the time, what you ate or were exposed to, the symptom and its severity from 0 to 10, and the minutes from exposure to symptom. The timing is the informative part. Within minutes points one direction, three to six hours points another, and next-day points somewhere else entirely. Do this for the whole trial, because the popular biomarkers do not work here and your own pattern really is the better data. Plasma histamine did not move at all in healthy volunteers who took 75 mg of oral histamine and developed real symptoms from it (PMID 15603203).

Change nothing else. No new supplements during the elimination phase. This is the same principle as the supplement guide's two-week gate, run in the other direction: an elimination phase underneath a changing supplement list cannot read out either.

Weeks 4 onward, the reintroduction, which is the actual test. One food at a time, a real portion, on a day you feel baseline, with three days before the next one. Write down what happened.

Reintroduction is where the information is. Elimination tells you that you feel better on a narrower diet, which given the placebo rate is weak evidence. Reintroduction tells you which specific things reproduce your symptoms, which is much stronger.

Track breadth monthly. Count the number of distinct foods you eat in a normal week. That number should be going up from month two onward. If it is going down, come back and we will change the plan.

Realistic expectations

Some people get most of the way home in three weeks on the storage rules alone. Others get a clear partial response and then plateau, which usually means the load side genuinely contributes and there is a second mechanism underneath it. A third group gets nothing at all, and that result is genuinely useful because it points away from histamine and toward the other mediator arms.

What I would not expect: that the diet fixes the mediator problem in someone whose picture is activation-dominant. That result reflects the diet acting on a compartment that was never the driver, not a failure of your discipline.

And I would not expect the diet to change your enzyme. It did not change diamine oxidase activity in any of the three trials that measured it (PMID 27624921, PMID 29606813, PMID 38769188).

When to seek clinical review rather than more restriction

- Any reaction with breathing difficulty, throat or tongue swelling, or fainting. That is emergency care, and then a physician conversation about an epinephrine prescription.
- Unintended weight loss, or a diet that has narrowed to fewer than roughly twenty foods.
- New or worsening symptoms during an elimination phase, which is information rather than progress.
- Iron deficiency, heavy menstrual bleeding, or persistent exhaustion. Those have their own workup and they outrank the food list.
- No response at all after a well-run three week trial, because that is a signal to change levers rather than to restrict harder.
- Any plan you have been on for more than three months with no reintroduction phase.

How you eat, which matters more than one more removed food

Three habits do more than the last ten items removed from your list.

Cook fresh and eat it. The single change with the best return in this whole guide.

Freeze rather than refrigerate. Same principle, applied to the food you cannot eat immediately.

Eat enough, and eat variety. A person on a narrow diet who is also undereating has two problems, and the second one is doing more damage. Dietary breadth is a clinical target here, not a luxury.

Working with Root Health

If you want help deciding which lever applies to you, or you have been on a restricted diet longer than you meant to be, the next step is a conversation. The Root Discovery Session is free and exists for exactly that. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere. In this condition it does real work, because the timed pattern it captures outperforms most of the testing available to you.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

This guide is educational. It is not medical advice, diagnosis, or treatment, and reading it does not create a doctor and patient relationship.

References

1. Placebo-controlled histamine challenge disproves suspicion of histamine intolerance. *J Allergy Clin Immunol Pract.* 2023. PMID 37648152.
2. A popular myth: low-histamine diet improves chronic spontaneous urticaria, fact or fiction? *J Eur Acad Dermatol Venereol.* 2017. PMID 27624921.
3. Son JH, et al. A histamine-free diet is helpful for treatment of adult patients with chronic spontaneous urticaria. *Ann Dermatol.* 2018. PMID 29606813.
4. Measurement of diamine oxidase during low-histamine or ordinary diet in patients with histamine intolerance. *Eur J Clin Nutr.* 2024. PMID 38769188.
5. Feng C, et al. Histamine (scombroid) fish poisoning: a comprehensive review. *Clin Rev Allergy Immunol.* 2016. PMID 25876709.
6. Emborg J, Dalgaard P. Significant histamine formation in tuna at 2 degrees C: effect of vacuum and modified atmosphere packaging. *Int J Food Microbiol.* 2005. PMID 15925710.
7. Growth, inactivation and histamine formation of *Morganella psychrotolerans* and *Morganella morganii*. *Int J Food Microbiol.* 2008. PMID 18845350.
8. The rate of histamine degradation by diamine oxidase is compromised by other biogenic amines. *Front Nutr.* 2022. PMID 35694170.
9. Diamine oxidase deficiency: implications for health, current management, and future directions. *Int J Biol Macromol.* 2025. PMID 40865824.
10. Sanchez-Perez S, et al. Low-histamine diets: is the exclusion of foods justified by their histamine content? *Nutrients.* 2021. PMID 33919293.
11. De Palma G, et al. Histamine production by the gut microbiota induces visceral hyperalgesia through histamine 4 receptor signaling in mice. *Sci Transl Med.* 2022. PMID 35895832.
12. Sanchez-Perez S, et al. Intestinal dysbiosis in patients with histamine intolerance. *Nutrients.* 2022. PMID 35565742.
13. Schink M, et al. Microbial patterns in patients with histamine intolerance. *J Physiol Pharmacol.* 2018. PMID 30552302.
14. McIntosh K, et al. FODMAPs alter symptoms and the metabolome of patients with IBS: a randomised controlled trial. *Gut.* 2017. PMID 26976734.
15. Wastyk HC, et al. Gut-microbiota-targeted diets modulate human immune status. *Cell.* 2021. PMID 34256014.
16. Recombinant human diamine oxidase activity is not inhibited by ethanol, acetaldehyde, disulfiram, diethyldithiocarbamate. *Alcohol.* 2016. PMID 27401969.
17. McCabe BJ. Dietary tyramine and other pressor amines in MAOI regimens: a review. *J Am Diet Assoc.* 1986. PMID 3525654.
18. Azzaro AJ, et al. Effects of a tyramine-enriched meal on blood pressure response in healthy male volunteers treated with selegiline transdermal system. *CNS Spectr.* 2007. PMID 17192761.
19. Diamine oxidase interactions with anti-inflammatory and anti-migraine medicines in the treatment of migraine. *J Clin Med.* 2023. PMID 38068554.

20. Wollin A, et al. Postheparin plasma diamine oxidase in health and intestinal disease. *Gastroenterology*. 1990. PMID 2110916.
21. Wohrl S, et al. Histamine intolerance-like symptoms in healthy volunteers after oral provocation with liquid histamine. *Allergy Asthma Proc*. 2004. PMID 15603203.

Read this part first

This guide is educational. It is not medical advice, it does not diagnose mast cell activation syndrome or histamine intolerance or anything else, and reading it does not make you my patient.

Anaphylaxis is a medical emergency. Difficulty breathing, throat or tongue swelling, a drop in blood pressure, fainting, or a reaction that keeps escalating after the trigger is gone means calling 911. If you have ever had a reaction like that, the most important thing in your care is an emergency action plan and an epinephrine prescription from a physician. Nothing in this guide replaces either one or should delay getting them.

Symptoms attributed to mast cells and histamine also come from true allergy, asthma, carcinoid and other neuroendocrine tumors, pheochromocytoma, thyroid disease, hereditary angioedema, autonomic disorders, anemia, infection, and medication reactions. New, severe, rapidly worsening, or unexplained symptoms get evaluated, not supplemented. Ruling those out is the foundation this entire guide sits on, not a delay tactic.

Pregnancy, breastfeeding, kidney disease, liver disease, bleeding disorders, and any prescription medication change what is safe here. Talk to the clinician who manages that part of your care before you start.

What you are buying, stated plainly

Almost every trial cited in this guide was run in systemic mastocytosis, chronic spontaneous urticaria, allergic rhinitis, migraine, irritable bowel syndrome, histamine intolerance, or healthy volunteers. Very little of it was run in a population defined the way you were probably defined.

What you are getting from me is established mechanism, trials in adjacent populations, and reasoned clinical practice held to a measurable endpoint. It is not a set of outcome trials in your exact diagnosis, because those do not exist.

I would rather tell you that on the first page than let you discover it later. It also means something useful: because the evidence is uneven, the order I give these things in is doing more work than any single item on the list.

The gate: get the diet stable before you start

This is the most important instruction in the guide, and it is the one most often skipped.

Hold a stable diet for two weeks before you introduce the first supplement.

Here is why, and it is not a preference. In fifty-nine patients with suspected histamine intolerance, a single-blind placebo-controlled oral histamine challenge excluded the diagnosis in fifty of them. Thirty-seven of the fifty-nine, about 63 percent, reacted to the placebo (PMID 37648152).

A 63 percent placebo response rate is not a footnote. It means that adding one agent every three to five days on top of a fluctuating dietary input will generate false positives and false negatives at a high rate. You will conclude you react to things you do not react to and you will discard things that were helping. And every decision made afterward inherits those errors.

Two weeks of a stable input is what makes the rest of this guide readable.

The cardinal rules that govern everything below

One agent at a time. Not two. Not a formula containing six things.

Fourteen days before you judge it. Long enough for a real effect, short enough not to waste a season.

Dilute it. This is specific to this condition and it is covered in its own section below. Concentrated solutions trigger mast cells directly, in a dose-dependent way, through osmolality alone (PMID 2449834, PMID 31207273).

Start below the studied dose and work up. Half, then three quarters, then the target.

Write it down. Start date, form, dose, what you expected to change, what actually changed by day 14, kept or stopped.

Two hours away from binders, if you use them at all.

Never with iron. Quercetin, polyphenols, calcium, zinc and binders all belong away from any iron dose.

When you feel worse

Feeling worse is information, not proof of anything.

A framework where feeling worse means it is working cannot be wrong about anything, which is the problem with it. Sometimes an adjustment period passes in a few days. Sometimes the dose is too high, sometimes the form is wrong, sometimes the excipient is the problem, and sometimes the groundwork was skipped and the input was never stabilized.

In this condition specifically there is a fourth possibility that does not apply elsewhere: you may be reacting to the concentration rather than the compound. Before you conclude an agent is out, dilute it in far more water, take it with food, and halve the dose. A surprising number of “reactions” do not survive that.

Significant worsening means stop and reassess with me. A reaction with breathing difficulty, throat swelling, or fainting means emergency care, not a dose adjustment.

1. The clearance side: helping you get rid of what arrives

This section is for the person whose picture is food-timed, gut-dominant, and worse with leftovers and aged foods. It works on the enzyme that clears histamine outside your cells.

Before anything here, understand the shape of this system, because it saves money. Diamine oxidase behaves as a **floor, not a dial**. In a controlled study, a frankly copper deficient diet lowered the enzyme, while supplementing copper at roughly five times adequate intake changed nothing at all (PMID 10953681).

There are exactly four ways to move it. Repair the gut lining. Correct a real copper deficiency. Put enzyme in the lumen. Reduce what arrives. Everything else sold for “DAO support” is acting on a system that does not work that way.

Oral diamine oxidase enzyme, before amine-containing meals

What it does. Swallowed DAO acts in the gut lumen on histamine arriving from food, as a substrate-clearing tool at the point of entry. It is not absorbed and it does not raise your own enzyme levels.

Dose and timing. The studied protocols give it roughly fifteen minutes before a meal, at the manufacturer's stated activity units. Porcine kidney extract is the form used in the trials.

The evidence, honestly. Two randomized double-blind placebo-controlled trials in adjacent populations (PMID 29698966, PMID 29475774), a genotype-selected trial in insomnia symptoms (PMID 41437955), and an open-label withdrawal study (PMID 31807350). The limits are that all were small and short, most were conducted close to industry, and in the migraine trial the between-group difference did not reach statistical significance.

The consistent signal across all of them is that benefit concentrates in people selected for **low baseline serum DAO**. That is why this is a targeted trial rather than a default purchase.

Who I use it in. Someone with a food-timed pattern, a low or low-normal serum DAO, and ideally an AOC1 variant. Run for four weeks with a stated endpoint. If nothing changes, stop. It is not a supplement you take forever on faith.

Not for. Anyone whose antihistamine trial did nothing, because that person's problem is probably not histamine at all.

Copper, at adequacy, and usually because zinc caused the problem

What it does. Copper is required by diamine oxidase twice over, once to build the enzyme's catalytic machinery and again to run it (PMID 20722416, PMID 27063138).

Dose. 1 to 2 mg per day, and only against a measured deficiency or a clear zinc excess. Not blind, and not above adequacy, because above adequacy does nothing (PMID 10953681).

The real-world version of this. The copper deficiency I actually find is almost always induced by someone taking 50 mg of zinc daily for a year. Fixing that is usually a matter of lowering the zinc, not adding copper indefinitely.

Test first. Serum zinc, serum copper and ceruloplasmin together, per the lab guide.

Vitamin C, at the dose that was actually studied

What it does. Vitamin C has a real place at this node, and it has a specific dose attached to it.

Dose. 2 g per day. This is a correction worth stating plainly: doses in the 250 to 500 mg range sit below every dose that has ever shown an effect at this node, which means that as written they do nothing for the reason they were added.

Caution. High-dose vitamin C loosens stools in some people and is best divided. It also enhances iron absorption, which is useful when you want that and worth knowing when you do not.

Gut lining repair, which is the actual lever

This piece gets the least attention and does the most work.

DAO capacity is essentially a function of intact small intestinal villous surface. Post-heparin plasma DAO correlates with jejunal mucosal DAO at r equal to 0.84, drops in active Crohn's disease in proportion to disease activity, and recovers in remission (PMID 2110916). In celiac disease, post-heparin DAO rose into the normal range within three months of a gluten free diet as the villi recovered (PMID 3121453).

Two consequences. First, an undiagnosed mucosal disease is a hard ceiling on your clearance capacity and no capsule lifts it. Get the celiac screen. Second, gut repair work is not a side quest here; it is the main intervention on this enzyme.

And the caution that runs the other way. Post-heparin DAO was 4.5 during total parenteral nutrition against 35.9 in controls, and reversed on resumption of oral feeding (PMID 2110916). Food in the lumen maintains the cell mass that **is** the enzyme.

Intravenous feeding is a far more extreme insult than an elimination diet and I am not going to overstate this. But the direction should concern anyone who has been eating fifteen foods for a year. A prolonged, very narrow diet may be eroding the clearance capacity being blamed for the problem. This is why every restriction in my plans carries an end date and a floor on dietary breadth.

Probiotics, where the strain decides everything

The problem. Some bacteria produce histamine. That is not a fringe concern: fecal bacteria from IBS patients with high urinary histamine produce large amounts of it in the laboratory, and a specific organism carrying a histamine-producing gene variant was over-represented across three independent human cohorts (PMID 35895832). Case-control work in histamine intolerance finds a consistent dysbiosis pattern (PMID 35565742, PMID 30552302).

What follows. A probiotic in this population has to be chosen by named strain, screened against histamine production, not by colony count and not by brand. The bacterial enzymes that make histamine come in two structurally unrelated families, one of which does not use vitamin B6 at all, which is why no nutrient switches them off (PMID 18756395, PMID 2857718, PMID 3082865).

What I do. Name the strain or do not use it. A product that lists only genus and species is not enough information to act on here.

2. The activation side: calming the cell

This section is for the person with multi-system symptoms, physical and non-food triggers, and reactions to medications and excipients.

Palmitoylethanolamide, which is the one with the best human data

What it does. It is a fatty acid amide your body already makes as a local brake on mast cell activation. Impaired release of this exact compound in the duodenum is what underlies acid-induced mast cell activation in human functional dyspepsia (PMID 33065341), which is unusually direct mechanistic grounding for a supplement.

Dose. 350 mg per day of the form used in the trial, a dispersible palmitoylethanolamide preparation. Higher ultramicrosized doses have been used in other settings.

The evidence. A 108-person double-blind randomized placebo-controlled trial found 350 mg per day for fourteen days significantly lowered histamine, interleukin 4, interleukin 8, interleukin 10 and TNF-alpha versus placebo (PMID 38068797).

The caveat that travels with it permanently. The primary symptom endpoint in that trial was negative overall and positive only in the moderate-severity subgroup. I am not going to quote the biomarker result without that sentence attached.

Even so: this is better human evidence than most of what is sold in this category, and it is the first thing I reach for on this node.

Why the form matters. Palmitoylethanolamide is poorly soluble. Dispersible and ultramicrosized preparations exist precisely because the raw powder absorbs badly.

Vitamin D, tested and to target

What it does. The vitamin D receptor physically blocks a key mast cell activation signaling step, and mast cells spontaneously activate in a vitamin D deficient environment (PMID 27998003).

Dose. Whatever brings your 25-OH vitamin D into 40 to 80 ng/mL, which means testing first and retesting after. A dose without a target is unauditabile, and this is a node where the target is the point.

The evidence. Mechanism above, plus randomized support as an add-on to antihistamine in chronic urticaria (PMID 26097642).

Form and cofactors. D3 with K2, taken with fat, in the smallest number of milliliters of oil that delivers the dose.

Quercetin, in the form that was actually studied

The correction that matters most here. The two positive randomized trials used **bioavailable** forms at 200 mg per day: enzymatically modified isoquercitrin in one (PMID 19295240) and a quercetin-containing supplement in the other (PMID 35776034).

Plain quercetin aglycone is absorbed poorly enough that 500 mg twice daily of it can deliver less active compound than 200 mg of a good form. Form beats dose here, which is the opposite of most people's instinct.

Dose. 200 mg per day of a bioavailable form.

Two honest limits. Quercetin has no evidence at the DAO clearance node despite being marketed for it, so it is a stabilization agent and not a clearance agent. And it inhibits iron absorption, so it never shares a dose with iron or an iron-rich meal.

Luteolin, labeled for what it is

What it does. Structurally related to quercetin, widely promoted in this space.

The evidence. Animal and cell work only. There is no human trial measuring any mast cell endpoint. I use it riding alongside palmitoylethanolamide rather than as a standalone, and I label it low-tier because that is where the evidence puts it.

Butterbur, with a split verdict I will not smooth over

Where it works. A standardized, pyrrolizidine-alkaloid-free Petasites extract is the best-evidenced botanical in this whole guide for allergic rhinitis symptoms: equivalent to cetirizine in a randomized trial (PMID 11799030), dose-dependent in a multicentre placebo-controlled trial (PMID 15611396), and it attenuates the histamine-induced skin response in humans (PMID 14989395).

Where it does not. When the same extract was tested specifically against intestinal histamine handling, it changed neither DAO protein nor DAO activity, and at high concentrations it increased histamine flux across the barrier (PMID 35149385).

So. Reasonable for upper airway symptoms. Not a DAO agent, and I will not sell it as one. And only ever in a certified alkaloid-free preparation, because unpurified butterbur is hepatotoxic. This is not optional caution.

Boswellia, for the leukotriene arm, at modest confidence

Why it is here. If your urine panel is leukotriene-dominant, the supplement options are thin, and this is the only botanical with a randomized placebo-controlled trial built on 5-lipoxygenase inhibition (PMID 29210124).

The limits. Small, industry-run, and a two-herb combination rather than Boswellia alone. I hold it at modest confidence and I say so.

Dose. The trial used 200 mg per day of the standardized combination.

Magnesium, kept for its real jobs

Magnesium stays in most of my plans for sleep, muscle, bowel regularity and blood sugar. There is **no** human dose studied for a mast cell endpoint and I am not going to imply one. Glycinate or bisglycinate form, 200 to 400 mg of elemental magnesium per day.

Omega-3, with the null result quoted

The finding people leave out. Four grams of EPA plus two grams of DHA daily for three weeks did not change bronchial hyperresponsiveness or urinary mast cell mediator excretion in asthmatics, despite confirmed adherence (PMID 25321659).

I still use omega-3 fats in plenty of people for cardiovascular and inflammatory reasons. I do not file them under mast cell stabilization, because the one human trial that measured that endpoint was negative. Rationale honesty matters more than shelf placement.

One safety note that belongs here rather than elsewhere. High-dose fish oil is mildly antiplatelet, and mast cell granules release biologically active heparin, with documented clotting abnormalities in mast cell disease (PMID 26447996, PMID 39557214, PMID 34445665). The mastocytosis coagulopathy is documented; clinically meaningful additive bleeding risk from typical supplement doses is not. Low probability, high consequence, so review it before surgery or a dental procedure and tell the proceduralist about the mast cell picture.

3. The methylation exit

This section serves the intracellular route, which is the one that governs brain and airway symptoms and which diet and swallowed enzymes do not touch (PMID 7943261, PMID 27679412).

Read it with a clear head about the limits. The enzyme uses SAM as its methyl donor and leaves SAH behind, and SAH competitively inhibits it, so the variable that governs this route is the SAM to SAH ratio, not homocysteine on its own (PMID 7188427). The only validated methylation index is that ratio, and only a 2025 randomized trial has shown it can be moved with nutrition at all (PMID 40883125). Nobody can currently promise you a measurable result at this node, and I will not.

What is defensible is narrower and worth doing.

Riboflavin

The evidence. Riboflavin at 1.6 mg per day for sixteen weeks raised SAM specifically in people with the MTHFR 677TT genotype in a randomized trial (PMID 32330571). That is the cleanest human demonstration that a B vitamin can move the variable this enzyme depends on.

A second job. Riboflavin is the source of the FAD cofactor for monoamine oxidase B, which performs the second half of the intracellular clearance route: the methylated histamine product still has to be oxidized before it is excreted. Riboflavin is doing two things at this node, not one.

Dose. Ordinary doses. This is not a high-dose intervention.

B12, folate, B6 and betaine, as the four exits from homocysteine

Homocysteine has three exits, and SAH only clears if homocysteine keeps moving. Remethylation by methionine synthase needs methylcobalamin and 5-methyltetrahydrofolate (PMID 9398304). Remethylation by betaine-homocysteine methyltransferase needs betaine (PMID 9681996). Transsulfuration needs pyridoxal 5-phosphate.

Dose. B12 at 2000 mcg, given alongside folate, B6 and trimethylglycine rather than alone. Trimethylglycine, which is betaine, has meta-analytic evidence for lowering plasma homocysteine at 4 g per day or more over six weeks or more in healthy adults (PMID 23997720, PMID 12399266). It is the route still standing when methylfolate and methyl-B12 are not tolerated, which happens in this population more than most.

Test, do not guess. Homocysteine target is 5 to 7 umol/L, and lower is not better. A homocysteine driven very low signals overmethylation and B vitamin intolerance.

The competition worth knowing about. HNMT is a small consumer of SAM in a crowded pool. Creatine synthesis, phospholipid methylation and nicotinamide clearance are all far larger sinks (PMID 9597750, PMID 26598833, PMID 28291578, PMID 24717514). That competition is real biology. It is also not measurable in a way that would let anyone build a protocol on it, so I am telling you it exists and stopping there.

4. What I do not use here, and why

Rule of thumb: the reason an agent is absent from this guide is usually that it was tested at this node and did not work, not that nobody thought of it.

The glutathione block

N-acetylcysteine, liposomal or acetyl glutathione, selenium, alpha lipoic acid and milk thistle have **no human study** against any mast cell or diamine oxidase endpoint. There is a mechanistic wrinkle, which is that the enzyme making prostaglandin D2 is glutathione dependent (PMID 12895599) and the enzyme making LTC4 conjugates glutathione directly. I am not going to build a claim on that in either direction.

These agents have a real place in my practice. That place is the chemical sensitivity and detoxification work, where they belong, not here.

High-dose zinc as a stabilizer

This one is worth reading twice, because it is backwards in most protocols. **Chelating** zinc blocks IgE-induced mast cell degranulation, and human mast cells actively release zinc when their receptors aggregate (PMID 16818790, PMID 23711853). Zinc is permissive **for** degranulation, not protective against it.

Zinc has real uses. Piling it on as a mast cell stabilizer is not one of them, and at sustained high doses it costs you the copper your clearance enzyme requires.

Binders

Activated charcoal, bentonite and chlorella have no evidence at any node in this system, and they carry a real cost: they bind minerals and medication indiscriminately, including the iron this population frequently needs. If they are used at all, they go two hours away from everything else.

Vitamin B6 for “DAO support”

Diamine oxidase is a copper and topaquinone enzyme, not a pyridoxal-phosphate enzyme (PMID 20722416, PMID 27063138). The B6-dependent enzyme in this system is histidine decarboxylase, the one that **makes** histamine (PMID 413882, PMID 7075603).

This is the single most common error in consumer DAO marketing, and it points the wrong way. B6 has other reasons to be in a plan. Supporting DAO is not one of them.

Anything sold on the DAO-inhibiting drug list

The circulating lists are largely unsupported. Tested directly, ibuprofen, aspirin, paracetamol, sumatriptan and zolmitriptan did not inhibit DAO. Only naproxen lowered DAO protein, without affecting activity (PMID 38068554). Ethanol, acetaldehyde and disulfiram do not inhibit human DAO either, and the apparent disulfiram effect was an assay artifact (PMID 27401969).

Alcohol still earns a conversation, on amine content and mast cell provocation. Not on DAO.

5. The supplement-specific hazards nobody warns you about

This section exists because in this condition the delivery vehicle can cause the reaction, and people spend months blaming ingredients.

Osmolality

Concentrated solutions trigger mast cells directly and dose-dependently. Nasal instillation of a hyperosmolar solution in healthy volunteers released histamine and leukotrienes compared with an isosmolar control, with more osmolality producing more histamine (PMID 2449834). The same stimulus produces mast cell dependent contraction in isolated human bronchi (PMID 31207273).

Now apply that to a protocol. Concentrated powders in a few ounces of water, undiluted liquid supplements, hypertonic nasal rinses and osmotic laxatives all raise local osmolality directly.

What to do. Dilute everything more than feels necessary. Take supplements with food and with a full glass of water. This costs nothing and it resolves a meaningful share of “reactions.”

Excipients

Mast cells carry a receptor, MRGPRX2, that responds to a whole class of compounds with no IgE involvement at all (PMID 25517090). Polysorbate 80, polyethylene glycol and certain other excipients have challenge-confirmed reactions behind them.

What to do. When you react to a product, change the form before you discard the ingredient. A different manufacturer, a powder instead of a softgel, a capsule opened into food. If the reaction does not follow the ingredient across forms, it was never the ingredient.

Speed

Three explanations sit in front of “I react to this compound,” and all three are cheaper to test. The excipient. The concentration. The speed of introduction.

Halve the dose, double the water, slow down, and see whether the reaction survives all three.

6. If the iron branch applies to you

A large share of people in this space are exhausted, foggy and dizzy on standing, and have attributed all of it to mediator burden. Some of it is. A great deal of it is iron, and iron is the most fixable thing in this guide.

Test properly. Ferritin, transferrin saturation and CRP together, never ferritin alone, because inflammation drives hepcidin and produces iron-restricted red cell production with a normal or high ferritin (PMID 19786207).

Hemoglobin is the wrong screening test. Fatigue responds to iron **without** anemia, with a pooled effect size of 0.33 and zero heterogeneity across six randomized trials (PMID 28625177), and the Cochrane review of 67 trials in 8,506 menstruating women found iron reduced anemia, raised hemoglobin, improved exercise performance and reduced fatigue (PMID 27087396).

Address the source, not only the level. Absorptive replacement tops out near 1 to 2 mg per day, so as long as monthly loss exceeds that ceiling, repletion is temporary. Heavy menstrual bleeding is a gynecological conversation with real options, and von Willebrand disease and structural causes outrank mast cells here by a wide margin (PMID 40697484, PMID 40955497).

Check the drug that may be causing it. Long-term H2 blockers and proton pump inhibitors carry odds ratios of 1.58 and 2.49 for iron deficiency, reaching 4.27 at high long-term use, and 1.25 and 1.65 for B12 deficiency (PMID 27890768, PMID 24327038). This is the backbone of conventional mast cell management, so it is worth monitoring rather than assuming.

Timing rules for repletion. Quercetin, polyphenols, calcium, zinc, silymarin and binders all go away from any iron dose and away from iron-rich meals. Riboflavin and vitamin A alongside iron reduced anemia prevalence in a randomized trial, which is a reminder that iron does not work alone (PMID 18806105).

The order I would actually run this in

If you take nothing else from this guide, take the sequence.

Weeks 1 and 2. Diet stable. Baselines drawn. Nothing new started. The medication screen done.

Weeks 3 and 4. One agent. Whichever your pattern points at: the enzyme if you are food-timed with a low DAO, palmitoylethanolamide if you are activation-dominant, iron if the depletion pattern is present. Diluted, below the target dose, working up.

Weeks 5 through 8. The second agent, on the same rules. Vitamin D correction runs in the background because it is dose-to-target rather than trial-and-error.

Weeks 9 through 12. Phenotype-directed work, once the urine panel has told you which arm is driving, and the prescription conversation with the physician who owns it.

Throughout. Reintroduction. Every restriction has an end date. The number of foods you can eat should be going up, and if it is going down, something in the plan is wrong.

Where to find these, and what I earn

I hold no physical inventory. I keep a screened dispensary through Fullscript, which ships directly to you, at us.fullscript.com/welcome/drseay. **Root Health earns a portion of dispensary purchases.** That is a real financial interest and you should know about it when you read a recommendation from me.

Where I have named a specific form rather than a brand, that is deliberate. In this category the form is what carries the evidence, whether that means a bioavailable quercetin, a dispersible palmitoylethanolamide, or a certified alkaloid-free butterbur. Anything I name that is not carried on Fullscript is available from any reputable retailer, and I would rather you got the right form than the one with my link on it.

Working with Root Health

If you want help deciding which of these applies to you and in what order, the next step is a conversation. The Root Discovery Session is free and exists for exactly that. Book at rootcauseunlocked.com/discovery.

Consultations and ongoing care are both available wherever you are located. Being outside North Carolina is not a barrier to working with me. Tell me where you are when you reach out and I will lay out exactly how we would work together from there.

If you would rather start on your own, the symptom tracker at rootcauseunlocked.com/tracker is free and works anywhere. Given the placebo response rate in this condition, tracking is far more than optional decoration. It is the instrument that tells you whether any of this is working.

Reach me at (919) 662-0520 or at 503 US-70 East, Suite L, Garner, NC 27529.

This guide is educational. It is not medical advice, diagnosis, or treatment, and reading it does not create a doctor and patient relationship.

References

1. Placebo-controlled histamine challenge disproves suspicion of histamine intolerance. *J Allergy Clin Immunol Pract.* 2023. PMID 37648152.
2. Eggleston PA, et al. In vivo release of inflammatory mediators by hyperosmolar solutions. *Am Rev Respir Dis.* 1988. PMID 2449834.
3. Larsson J, et al. Mannitol triggers mast cell-dependent contractions of human small bronchi and prostacyclin bronchoprotection. *J Allergy Clin Immunol.* 2019. PMID 31207273.
4. Response of diamine oxidase and other plasma copper biomarkers to various dietary copper intakes in the rat. *Br J Nutr.* 2000. PMID 10953681.
5. Yacoub MR, et al. Diamine oxidase supplementation in chronic spontaneous urticaria: a randomized, double-blind placebo-controlled study. *Int Arch Allergy Immunol.* 2018. PMID 29698966.
6. Izquierdo-Casas J, et al. Diamine oxidase supplement reduces headache in episodic migraine patients with DAO deficiency: a randomized double-blind trial. *Clin Nutr.* 2019. PMID 29475774.
7. Effect of diamine oxidase enzyme dietary supplementation in subjects with insomnia symptoms and AOC1 single nucleotide polymorphisms. *Sleep Med X.* 2026. PMID 41437955.
8. Diamine oxidase supplementation improves symptoms in patients with histamine intolerance. *Food Sci Biotechnol.* 2019. PMID 31807350.
9. Correlation of active site metal content in human diamine oxidase with trihydroxyphenylalanine quinone cofactor biogenesis. *Biochemistry.* 2010. PMID 20722416.
10. Characterization of recombinant human diamine oxidase produced in CHO cells. *J Biotechnol.* 2016. PMID 27063138.
11. Wollin A, et al. Postheparin plasma diamine oxidase in health and intestinal disease. *Gastroenterology.* 1990. PMID 2110916.
12. Postheparin plasma diamine oxidase increases in patients with coeliac disease during gluten free diet. *Gut.* 1987. PMID 3121453.
13. De Palma G, et al. Histamine production by the gut microbiota induces visceral hyperalgesia through histamine 4 receptor signaling in mice. *Sci Transl Med.* 2022. PMID 35895832.
14. Sanchez-Perez S, et al. Intestinal dysbiosis in patients with histamine intolerance. *Nutrients.* 2022. PMID 35565742.
15. Schink M, et al. Microbial patterns in patients with histamine intolerance. *J Physiol Pharmacol.* 2018. PMID 30552302.
16. Landete JM, et al. Updated molecular knowledge about histamine biosynthesis by bacteria. *Crit Rev Food Sci Nutr.* 2008. PMID 18756395.
17. Pyruvoyl-dependent histidine decarboxylases: preparation and amino acid sequences. *J Biol Chem.* 1985. PMID 2857718.
18. Pyruvoyl-dependent histidine decarboxylase from *Lactobacillus* 30a: covalent modifications. *J Biol Chem.* 1986. PMID 3082865.
19. Wouters MM, et al. Impaired duodenal palmitoylethanolamide release underlies acid-induced mast cell activation in functional dyspepsia. *Cell Mol Gastroenterol Hepatol.* 2021. PMID 33065341.
20. Briskey D, et al. The effect of Levagen+ (palmitoylethanolamide) supplementation on symptoms of allergic rhinitis: a double-blind placebo-controlled study. *Nutrients.* 2023. PMID 38068797.

21. Liu ZQ, et al. Vitamin D contributes to mast cell stabilization. *Allergy*. 2017. PMID 27998003.
22. Rasool R, et al. Chronic urticaria merits serum vitamin D evaluation and supplementation: a randomized case control study. *World Allergy Organ J*. 2015. PMID 26097642.
23. Kawai M, et al. Effect of enzymatically modified isoquercitrin on symptoms of Japanese cedar pollinosis: a randomized double-blind placebo-controlled trial. *Int Arch Allergy Immunol*. 2009. PMID 19295240.
24. Effects of repeated oral intake of a quercetin-containing supplement on allergic reaction: a randomized, placebo-controlled trial. *Eur Rev Med Pharmacol Sci*. 2022. PMID 35776034.
25. Schapowal A. Randomised controlled trial of butterbur and cetirizine for treating seasonal allergic rhinitis. *BMJ*. 2002. PMID 11799030.
26. Schapowal A. Butterbur Ze339 for the treatment of intermittent allergic rhinitis: dose-dependent efficacy. *Arch Otolaryngol Head Neck Surg*. 2004. PMID 15611396.
27. Jackson CM, et al. The effects of butterbur on the histamine and allergen cutaneous response. *Ann Allergy Asthma Immunol*. 2004. PMID 14989395.
28. Impact of the clinically approved Petasites hybridus extract Ze 339 on intestinal mechanisms involved in the handling of histamine. *Biomed Pharmacother*. 2022. PMID 35149385.
29. A novel herbal composition containing extracts of Boswellia serrata gum resin and Aegle marmelos fruit alleviates symptoms of asthma. *Phytother Res*. 2018. PMID 29210124.
30. Brannan JD, et al. The effect of omega-3 fatty acids on bronchial hyperresponsiveness, sputum eosinophilia, and mast cell mediators in asthma. *Chest*. 2015. PMID 25321659.
31. Barete S, et al. A French national survey on clotting disorders in mastocytosis. *Medicine (Baltimore)*. 2015. PMID 26447996.
32. Thromboelastography for rapid diagnosis of heparin-like anticoagulant release during anaphylaxis-induced coagulopathy. *Clin Med (Lond)*. 2025. PMID 39557214.
33. Effects of primary mast cell disease on hemostasis and erythropoiesis. *Int J Mol Sci*. 2021. PMID 34445665.
34. Yamauchi K, et al. Structure and function of human histamine N-methyltransferase: critical enzyme in histamine metabolism in airway. *Am J Physiol*. 1994. PMID 7943261.
35. Histamine clearance through polyspecific transporters in the brain. *Handb Exp Pharmacol*. 2017. PMID 27679412.
36. Francis DM, et al. The kinetic properties and reaction mechanism of histamine methyltransferase from human skin. *Biochem J*. 1980. PMID 7188427.
37. Lowering plasma S-adenosylhomocysteine in healthy adults with elevated SAH and normal homocysteine using nutritional supplementation. *Nutr Metab Cardiovasc Dis*. 2025. PMID 40883125.
38. Impact of the MTHFR C677T polymorphism on one-carbon metabolites: evidence from a randomised trial of riboflavin supplementation. *Biochimie*. 2020. PMID 32330571.
39. Goulding CW, Matthews RG. Cobalamin-dependent methionine synthase: involvement of zinc in homocysteine activation. *Biochemistry*. 1997. PMID 9398304.
40. Millian NS, Garrow TA. Human betaine-homocysteine methyltransferase is a zinc metalloenzyme. *Arch Biochem Biophys*. 1998. PMID 9681996.
41. Olthof MR, et al. Betaine supplementation decreases plasma homocysteine in healthy adult participants: a meta-analysis. *J Chiropr Med*. 2013. PMID 23997720.

42. Betaine supplementation decreases plasma homocysteine concentrations but does not affect body weight or body composition. *Am J Clin Nutr*. 2002. PMID 12399266.
43. Yeo EJ, Wagner C. Structure, function and physiological role of glycine N-methyltransferase. *Int J Biochem Cell Biol*. 1998. PMID 9597750.
44. Regulation of folate-mediated one-carbon metabolism by glycine N-methyltransferase and MTHFR. *J Nutr Sci Vitaminol (Tokyo)*. 2015. PMID 26598833.
45. Pissios P. Nicotinamide N-methyltransferase: more than a vitamin B3 clearance enzyme. *Trends Endocrinol Metab*. 2017. PMID 28291578.
46. Kraus D, et al. Nicotinamide N-methyltransferase knockdown protects against diet-induced obesity. *Nature*. 2014. PMID 24717514.
47. Kanaoka Y, Urade Y. Hematopoietic prostaglandin D synthase. *Prostaglandins Leukot Essent Fatty Acids*. 2003. PMID 12895599.
48. Kabu K, et al. Zinc is required for FcεRI-mediated mast cell activation. *J Immunol*. 2006. PMID 16818790.
49. Regulation of IgE-dependent zinc release from human mast cells. *Int Arch Allergy Immunol*. 2013. PMID 23711853.
50. Pyridoxal-5'-phosphate as a cofactor for rat brain histidine decarboxylase. *J Neurochem*. 1978. PMID 413882.
51. Properties of histidine decarboxylase from rat gastric mucosa. *Eur J Biochem*. 1982. PMID 7075603.
52. Diamine oxidase interactions with anti-inflammatory and anti-migraine medicines in the treatment of migraine. *J Clin Med*. 2023. PMID 38068554.
53. Recombinant human diamine oxidase activity is not inhibited by ethanol, acetaldehyde, disulfiram, diethyldithiocarbamate. *Alcohol*. 2016. PMID 27401969.
54. McNeil BD, et al. Identification of a mast-cell-specific receptor crucial for pseudo-allergic drug reactions. *Nature*. 2015. PMID 25517090.
55. Ganz T, Nemeth E. Iron sequestration and anemia of inflammation. *Semin Hematol*. 2009. PMID 19786207.
56. Houston BL, et al. Iron deficiency without anaemia is a potential cause of fatigue: meta-analyses of randomised controlled trials. *Br J Nutr*. 2017. PMID 28625177.
57. Low MSY, et al. Daily iron supplementation for improving anaemia, iron status and health in menstruating women. *Cochrane Database Syst Rev*. 2016. PMID 27087396.
58. Prevalence of heavy menstrual bleeding, iron deficiency, iron deficiency anemia, and treatment in women with von Willebrand disease. *Res Pract Thromb Haemost*. 2025. PMID 40697484.
59. Prevalence and impact of heavy menstrual bleeding in women with von Willebrand disease across age groups. *Haemophilia*. 2025. PMID 40955497.
60. Tran-Duy A, et al. Proton pump inhibitor and histamine-2 receptor antagonist use and iron deficiency. *Gastroenterology*. 2017. PMID 27890768.
61. Lam JR, et al. Proton pump inhibitor and histamine 2 receptor antagonist use and vitamin B12 deficiency. *JAMA*. 2013. PMID 24327038.
62. Retinol and riboflavin supplementation decreases the prevalence of anemia in Chinese pregnant women taking iron and folic acid. *J Nutr*. 2008. PMID 18806105.